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Theoretical Physics

Advanced Quantum Field Theory

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Introduction

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1 | Gauge Theories

Gauge theories are building blocks of the Standard Model of particle physics. They arise from the requirement that massless spin-one particles, which mediate some of the fundamental forces of nature, should carry only two independent polarizations¹ even when described in terms of equations of motion that are manifestly Lorentz invariant: the photon is conveniently described by the vector field $A_\mu(x)$, which has four independent components as any four-vector, though the expected physical degrees of freedom are only two. The other two degrees of freedom are redundant because of the gauge symmetry: one can eliminate them by performing gauge transformations to reach a unitary gauge, such as the Coulomb gauge used in electrodynamics. However, in such a gauge, the Lorentz symmetry is not manifest, and it becomes difficult to exploit its consequences in a useful manner. It is sometimes more convenient to keep some redundancy associated with the gauge symmetry and maintain the Lorentz invariance as manifest as possible. Then, the principle of gauge invariance can be used to fix in a simple way all possible interactions mediated by massless particles of spin one in a way that is consistent with Lorentz invariance. Here, we introduce the principle of gauge invariance and use it to streamline the construction of the QED and QCD lagrangians, the main examples of abelian and non-abelian gauge theories, respectively.

¹As implied by Wigner's classification of the irreducible unitary representations of the Poincaré group.

1.1 | Abelian gauge theories and QED

1.1.1 | The Free Dirac Field and Global Symmetry

Let us first review how, starting from the theory of free electrons described by the free Dirac equation, one elegantly deduces the complete Quantum Electrodynamics (QED) lagrangian. This derivation relies heavily on the principle of gauge invariance and the fundamental requirement of Lorentz invariance. We begin by considering the lagrangian of a free Dirac field of mass m :

$$\mathcal{L}_{Dirac} = -\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi \quad (1.1.1)$$

which is explicitly constructed to be Lorentz invariant, and thus perfectly consistent with special relativity.

Throughout this text, we employ a mostly plus metric convention, denoting the Minkowski metric by $\eta_{\mu\nu}$. The gamma matrices are taken to satisfy the fundamental Clifford algebra relation:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad (1.1.2)$$

while the conjugate Dirac spinor $\bar{\psi}$ is formally defined by:

$$\bar{\psi} = \psi^\dagger\beta, \quad \beta = i\gamma^0, \quad (\gamma^\mu)^\dagger = -\beta\gamma^\mu\beta. \quad (1.1.3)$$

One may straightforwardly check that the Dirac lagrangian is formally real up to total derivatives and, of course, Lorentz invariant. Crucially, it is also invariant under global internal symmetry transformations belonging to the Abelian group $U(1)$:

$$\begin{aligned} \psi(x) &\rightarrow \psi'(x) = e^{i\alpha}\psi(x), & e^{i\alpha} \in U(1) \\ \bar{\psi}(x) &\rightarrow \bar{\psi}'(x) = e^{-i\alpha}\bar{\psi}(x). \end{aligned} \quad (1.1.4)$$

This is classified as a global symmetry because the parameter α is a constant, entirely independent of spacetime coordinates.

1.1.2 | Local Gauge Invariance and the Covariant Derivative

Let us now explore how to extend this symmetry to a local one by introducing arbitrary spacetime-dependent functions $\alpha(x)$:

$$\psi(x) \rightarrow \psi'(x) = e^{i\alpha(x)}\psi(x) \quad (1.1.5)$$

$$\bar{\psi}(x) \rightarrow \bar{\psi}'(x) = e^{-i\alpha(x)}\bar{\psi}(x). \quad (1.1.6)$$

The mass term in the lagrangian is trivially invariant since the opposite phases cancel out perfectly:

$$m\bar{\psi}\psi \rightarrow m\bar{\psi}'\psi' = m\bar{\psi}e^{-i\alpha(x)}e^{i\alpha(x)}\psi = m\bar{\psi}\psi, \quad (1.1.7)$$

however, the term containing the partial derivative is not invariant:

$$\bar{\psi}\gamma^\mu\partial_\mu\psi \rightarrow \bar{\psi}'\gamma^\mu\partial_\mu\psi' = \bar{\psi}e^{-i\alpha(x)}\gamma^\mu\partial_\mu(e^{i\alpha(x)}\psi) = \bar{\psi}\gamma^\mu\partial_\mu\psi + i\bar{\psi}\gamma^\mu\psi\partial_\mu\alpha(x). \quad (1.1.8)$$

The local transformation produces an extra term that vanishes only for a constant $\alpha(x)$. Consequently, the free lagrangian is not invariant, and it must be modified to achieve gauge invariance—that is, invariance for arbitrary functions $\alpha(x)$. Incidentally, one may note that the term

multiplying the derivative of $\alpha(x)$ is directly proportional to the Noether current $J^\mu = i\bar{\psi}\gamma^\mu\psi$ associated with the global symmetry discussed in (1.1.4).

To rigorously construct gauge invariant actions, it is extremely useful to introduce a formalism based on the definition of *tensors* of the gauge group and their corresponding *covariant derivatives*. The latter are constructed specifically to produce tensors out of tensors, preserving the transformation properties.

To start with, we define $\psi(x)$ as a tensor under the gauge group $U(1) = \{e^{i\alpha(x)}\}$ if it transforms exactly as in (1.1.5). It follows that $\partial_\mu\psi(x)$ is not a tensor, as its transformation rule is more complicated (as seen above). The covariant derivative acting on ψ is obtained by introducing a new gauge field A_μ (often referred to in differential geometry as a “connection”). It is defined by:

$$D_\mu = \partial_\mu - iA_\mu(x). \quad (1.1.9)$$

To ensure the derivative behaves properly, one requires $A_\mu(x)$ to transform in a specific way so that the “tensorial” transformation rule remains valid for the derivative:

$$D_\mu\psi(x) \rightarrow D'_\mu\psi'(x) = e^{i\alpha(x)}D_\mu\psi(x) \quad (1.1.10)$$

where $D'_\mu = \partial_\mu - iA'_\mu(x)$. A short algebraic calculation demonstrates that the following transformation rule for the gauge field must hold to satisfy this requirement:

$$A_\mu(x) \rightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu\alpha(x). \quad (1.1.11)$$

Armed with covariant derivatives, it becomes simple to obtain a fully gauge invariant lagrangian from our starting point in (1.1.1):

$$\boxed{\mathcal{L} = -\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi}. \quad (1.1.12)$$

Comment: a “tensor” for the gauge group $U(1)$ is more generally defined as a field $\psi_q(x)$ that transforms as

$$\psi_q(x) \rightarrow \psi'_q(x) = e^{iq\alpha(x)}\psi_q(x) \quad (1.1.13)$$

where the integer $q \in \mathbb{Z}$ is called the “charge” of $\psi_q(x)$. Thus, $\psi_q(x)$ is considered a tensor of charge q : in more formal mathematical terms, q identifies a specific irreducible representation of the group $U(1)$. The general definition of the covariant derivative is then naturally extended to:

$$D_\mu = \partial_\mu - iA_\mu(x)Q \quad (1.1.14)$$

where Q is an operator that measures the charge of the tensor on which it acts, meaning it functions as the generator of the $U(1)$ group in the representation of the target tensor. We can now easily recognize that the transformation of the conjugate spinor in (1.1.6) corresponds precisely to that of a tensor of charge -1 , meaning the covariant derivative of $\bar{\psi}$ reads:

$$D_\mu\bar{\psi} = (\partial_\mu + iA_\mu(x))\bar{\psi}, \quad (1.1.15)$$

which is completely consistent with taking the Dirac conjugate of $D_\mu\psi$. The covariant derivative possesses the essential property that it does not destroy the tensorial character of the object on which it acts: *it generates tensors out of tensors*. Indeed, one can verify that

$$D_\mu\psi_q(x) = \partial_\mu\psi_q(x) - iqA_\mu(x)\psi_q(x) \quad (1.1.16)$$

is once again a tensor of charge q . Another highly useful property of this definition is the validity of the Leibniz rule for covariant derivatives: because products of tensors are also tensors, one can systematically check that

$$D_\mu(\psi_{q_1}\psi_{q_2}) = (D_\mu\psi_{q_1})\psi_{q_2} + \psi_{q_1}(D_\mu\psi_{q_2}). \quad (1.1.17)$$

1.1.3 | The Electromagnetic Field Tensor and Maxwell Lagrangian

We have demonstrated that local gauge invariance is successfully achieved by introducing the gauge field $A_\mu(x)$, which is readily interpreted physically as the potential of the electromagnetic field. However, having introduced a new dynamical field into the theory, one must grant it suitable dynamics by adding a kinetic term for $A_\mu(x)$ to the lagrangian. This new kinetic term must independently be gauge invariant because the rest of the Lagrangian already is: gauge symmetry remains the guiding principle for building the action.

It is conceptually useful to proceed using our established tensor formalism. We start by calculating the commutator of two covariant derivatives acting on the tensor ψ of charge 1:

$$[D_\mu, D_\nu]\psi = -iF_{\mu\nu}\psi \quad (1.1.18)$$

This relation defines the essential quantity $F_{\mu\nu}$. Since we have strictly tensorial quantities on the left-hand side, the right-hand side must mathematically also be built exclusively out of tensors. In this way, we deduce that $F_{\mu\nu}$ is a tensor of charge $q = 0$, meaning it is a quantity that is strictly invariant under gauge transformations (i.e., it satisfies a transformation with $q = 0$ in (1.1.13)). Computing the left-hand side of (1.1.18) explicitly, one finds

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (1.1.19)$$

which is readily recognized and interpreted as the electromagnetic field tensor. One can manually verify that it is indeed completely gauge invariant under the specific gauge transformation given in (1.1.11).

Now, it is mathematically straightforward to construct a gauge invariant lagrangian containing at most two derivatives on A_μ . It is sufficient to use as a basic building block the field strength $F_{\mu\nu}$, which being gauge invariant itself makes it incredibly easy to construct further gauge invariant quantities. One must concurrently require Lorentz invariance to maintain a relativistic theory, which ultimately leads to the free Maxwell lagrangian. In a standard normalization, this reads:

$$\mathcal{L}_{Maxwell} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (1.1.20)$$

1.1.4 | The Complete QED Lagrangian and Feynman Rules

Summing together all the individual pieces that are separately gauge invariant (i.e., combining (1.1.12) and (1.1.20)), one finally finds the full QED lagrangian:

$$\boxed{\mathcal{L}_{QED} = -\frac{1}{4e^2}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi} \quad (1.1.21)$$

where a free multiplicative parameter $1/e^2$ is introduced to account for a relative weight between the different terms that are separately gauge invariant.

Let us thoroughly analyze the various terms contained in (1.1.21). It is highly useful to redefine the gauge field via $A_\mu \rightarrow eA_\mu$ (doing so recovers the standard normalization of the free Maxwell action) and recognize that e is the physical coupling constant. Note that e now appears inside the

covariant derivative $D_\mu = \partial_\mu - ieA_\mu(x)$, taking $e < 0$ for the electron. Expanding this, we obtain:

$$\begin{aligned}\mathcal{L}_{QED} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu D_\mu + m)\psi \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \bar{\psi}(\gamma^\mu \partial_\mu + m)\psi + ieA_\mu \bar{\psi}\gamma^\mu \psi\end{aligned}\quad (1.1.22)$$

$$= \text{wavy line } \gamma + \text{solid line } e^- + \text{solid line } e^- \text{ with wavy line } \gamma \text{ attached} \quad (1.1.23)$$

The first term distinctly describes the free propagation of photons, the second one accounts for the free propagation of electrons, and the crucial third one dictates the elementary interaction between photons and electrons. The constant e is the coupling constant, appropriately identified with the elementary charge of the electron. Thus, the guiding gauge principle has powerfully allowed us to discover the exact interaction mechanism between fields of spin 1/2 and 1. Let us summarize once again the precise rules of gauge transformations for the lagrangian in (1.1.23): by rescaling for simplicity also the phase angle $\alpha(x) \rightarrow e\alpha(x)$, we have

$$\begin{aligned}\psi &\rightarrow \psi' = e^{ie\alpha}\psi \\ \bar{\psi} &\rightarrow \bar{\psi}' = e^{-ie\alpha}\bar{\psi} \\ A_\mu &\rightarrow A'_\mu = A_\mu + \partial_\mu \alpha.\end{aligned}\quad (1.1.24)$$

1.1.5 | Feynman Diagrams for Basic Scattering Processes

If the fundamental coupling constant e is sufficiently small, the theory can be successfully treated perturbatively, and the theoretical amplitudes for the various physical processes in QED can be systematically associated to Feynman diagrams constructed with the elementary interaction vertex found in (1.1.23). For example, the fundamental electron-electron scattering process (known as Möller scattering) at the lowest perturbative order is represented by:

where, by our adopted convention, time runs along the horizontal axis in our Feynman diagrams.

Other significant processes include electron-positron scattering (Bhabha scattering):

and the scattering of electrons and photons (Compton scattering):

Fascinatingly, pure photon-photon scattering is also dynamically possible. Because there is absolutely no elementary vertex coupling four photons directly, the first perturbative term that one calculates is given by the one-loop box graph:

together with topologically similar graphs where the external photon lines are logically attached to the internal vertices with completely different orderings. In general quantum field theories, loop corrections can be severely divergent and must be rigorously cured by the procedure of renormalization. However, quite remarkably, for photon-photon scattering, the total sum of the one-loop Feynman diagrams indicated above is ultraviolet finite, and strictly no four-photon counterterm is needed in the lagrangian. This elegant finiteness is a direct and beautiful consequence of the underlying gauge invariance of the theory.

TODO: add image of Moller scattering Feynman diagrams

TODO: add image of Bhabha scattering Feynman diagrams
TODO: add image of Compton scattering Feynman diagrams

TODO: add image of photon-photon scattering box diagram

1.2 | Non-abelian gauge theories and QCD

The theoretical framework used for the construction of gauge invariant actions can be elegantly extended to accommodate compact non-abelian groups. To accomplish this, we must first review the principal mathematical properties of Lie groups, leaving more detailed foundational considerations for a dedicated appendix.

1.2.1 | Lie groups

Before constructing non-abelian gauge theories, let us briefly review the characteristics of simple and compact Lie groups², keeping the special unitary group $SU(N)$ in mind as our primary illustrative example:

$$SU(N) = \{U \in GL(N, \mathbb{C}) \mid U^\dagger U = 1, \det U = 1\},$$

and we are interested to study $SU(N)$ since we can write the unitary group as a direct product of $U(1)$ and $SU(N)$:

$$U(N) = \cdots = U(1) \times SU(N).$$

An arbitrary element U of a compact, non-abelian Lie group G can be parameterized by continuous coordinates α_a (the group parameters), which are formally associated with hermitian generators T^a . A comprehensive list of the main definitions and properties is as follows:

- (i) $U = U(\alpha) = \exp(i\alpha_a T^a) \in G \quad a = 1, \dots, \dim G$,
- (ii) $[T^a, T^b] = if^{ab}_c T^c$,
- (iii) $\text{tr}(T^a_F T^b_F) = \frac{1}{2}\delta^{ab}$,
- (iv) $f^{abc} = f^{ab}_d \delta^{dc}$ is totally antisymmetric,
- (v) $[[T^a, T^b], T^c] + [[T^b, T^c], T^a] + [[T^c, T^a], T^b] = 0 \implies f^{ab}_d f^{dc}_e + f^{bc}_d f^{da}_e + f^{ca}_d f^{db}_e = 0$,
- (vi) $(T^a_{\text{Adj}})^b_c = -if^{ab}_c$.

To elaborate on these defining properties: Point (i) describes the exponential representation of an arbitrary element U of the group G that is continuously connected to the identity operator. This structure is verified by considering the defining representation, i.e., the specific matrix representation used to define the group. For $SU(N)$, this representation is given by unitary $N \times N$ matrices with a unit determinant. The index a takes as many values as the total dimension of the group. Therefore, an element of the group is parameterized by the continuous "angles" α_a that multiply the generators T^a . Matrices in the defining (also called fundamental) representation are explicitly written as T^a_F , but they are then interpreted as abstract generators T^a that may possess other representations.

²A *compact group* is defined as a group that is compact as a topological space. Intuitively, it can be conceptualized as a group possessing finite "size" or "volume". More precisely, a compact group has a finite volume with respect to a translation-invariant group theoretical measure known as the "Haar measure". A *simple Lie group* is, roughly speaking, a Lie group whose corresponding Lie algebra $\mathfrak{g}$ is simple. This condition indicates that $\mathfrak{g}$ is non-abelian and possesses no nontrivial ideals. An ideal $\mathfrak{i}$ is a subalgebra satisfying $[\mathfrak{g}, \mathfrak{i}] \subseteq \mathfrak{i}$. Thus, the only ideals of a simple Lie algebra are 0 and $\mathfrak{g}$ itself.

Point (ii) provides the **fundamental Lie algebra** satisfied by the hermitian generators T^a . The constants f_c^{ab} are strictly real and are known as the **structure constants** of the group. They uniquely characterize the group G and are manifestly antisymmetric in the initial indices a and b .

Point (iii) defines a standard convention for the normalization of the generators in the fundamental representation T_F^a . It formally identifies the so-called "*Killing metric*". More generally, one could have defined the Killing metric γ^{ab} by the trace relation $\text{tr}(T_F^a T_F^b) = \frac{1}{2}\gamma^{ab}$, and rigorously prove that the Killing metric must be positive definite for all compact Lie groups, such as $SU(N)$. The normalized basis chosen above for the generators T_F^a yields the Kronecker delta δ^{ab} as the Killing metric for the compact group G .

Point (iv) utilizes the established Killing metric to systematically raise the lower index of the structure constants. Following this operation, one can show that the resulting symbols f^{abc} are completely antisymmetric: It us prove this.

Proof. Antisymmetry on the indices a and b is intrinsically obvious from property (ii)...

Antisymmetry on the indices b and c is systematically deduced by multiplying the Lie algebra with an additional generator, taking the trace, and subsequently using point (iii) in conjunction with the cyclic property of the trace operation:

$$\begin{aligned} f^{abc} &= \dots \\ &= \dots \end{aligned}$$

explicitly proving the antisymmetry on the indices b and c . Antisymmetry on the indices a and c is then automatically implied by the previous two results. $\square$

Point (v) explicitly provides the **Jacobi identities** inherently satisfied by the structure constants.

Point (vi) defines the **adjoint representation**. It is formally proven to be a valid mathematical representation by utilizing the Jacobi identities. Note that the indices in $(T_{\text{Adj}}^a)^b_c$ all run from 1 to $\dim G$, so that the dimension of the adjoint representation naturally coincides with the dimension of the group itself, i.e. $\dim G$. In the specific case of $SU(N)$, these indices run from 1 to $N^2 - 1$, and the dimension of the adjoint representation is thus $N^2 - 1$.

1.2.2 | Action with rigid $SU(N)$ symmetry

Let us now introduce physical fields by considering N free Dirac fields, assembled structurally into column and row vectors as follows:

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \\ \vdots \\ \psi^N \end{pmatrix}, \quad \bar{\psi} = (\bar{\psi}_1, \bar{\psi}_2, \dots, \bar{\psi}_N),$$

which let us naturally study the fundamental representation of the group $SU(N)$. The fields are structured so that the corresponding scalar product

$$\bar{\psi}\psi = \bar{\psi}_1\psi^1 + \bar{\psi}_2\psi^2 + \dots + \bar{\psi}_N\psi^N,$$

is manifestly $SU(N)$ invariant. If these defined fields possess identical masses m , then there exists a global continuous symmetry. The free lagrangian is straightforwardly given by

$$\mathcal{L}_{Dirac} = -\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi, \tag{1.2.1}$$

and is invariant under the rigid $SU(N)$ symmetry transformations provided by

$$\begin{aligned}\psi(x) &\rightarrow \psi'(x) = U\psi(x), \\ \bar{\psi}(x) &\rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U^\dagger = \bar{\psi}(x)U^{-1},\end{aligned}\tag{1.2.2}$$

where $U \in SU(N)$, and $U^\dagger = U^{-1}$ inherently holds since U is a unitary matrix. These are classified as **global transformations**, as the group parameters α^a contained within $U = U(\alpha) = \exp(i\alpha^a T^a)$ are strictly constant across spacetime (note that indices in α^a are raised and lowered with the Killing metric, which coincides with the identity matrix in our chosen conventions).

It is critical to note that all the fermion masses must be strictly identical; otherwise, the mass term would dynamically break the $SU(N)$ symmetry. There is also an independent abelian $U(1)$ symmetry present, but the procedure for its gauging proceeds exactly as described previously.

1.2.3 | Covariant derivative

To introduce gauge interactions, we would like to mathematically extend the theory to achieve a local $SU(N)$ symmetry. For this purpose, it is once again convenient to introduce the powerful concept of **covariant derivatives**. By formal definition, the covariant derivative successfully produces new tensors when applied to existing tensors. To begin, we recall that a tensor $\psi(x)$ residing in the fundamental representation of the gauge group $SU(N)$, i.e., the representation usually indicated by its dimension $\mathbf{N}$, is a field defined by the transformation:

$$\psi(x) \rightarrow \psi'(x) = U(x)\psi(x),$$

where $U(x)$ is an $N \times N$ matrix of $SU(N)$ that can vary for any spacetime point x . More generally, fields transforming in any given mathematical representation $R(U(x))$ of the original matrices $U(x)$ are formally said to be tensors in the representation R . For example, the conjugate field $\bar{\psi}(x)$ is a tensor in the anti-fundamental representation, which is usually indicated by $\bar{\mathbf{N}}$ as it corresponds to the complex conjugate of the representation $\mathbf{N}$, and its transformation rule is written as follows:

$$\bar{\psi}(x) \rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U^{-1}(x).$$

Evidently, the bilinear term $\bar{\psi}(x)\psi(x)$ acts as a scalar under this local gauge transformation.

The covariant derivative acting on tensors produces new tensors. It is defined by the following operator:

$$D_\mu = \partial_\mu + W_\mu(x),\tag{1.2.3}$$

where $W_\mu(x)$ is a matrix-valued gauge field, which is also commonly known as the **connection** (geometrically, *it strictly defines a notion of parallel transport in a certain mathematical space*, known formally as a fiber bundle). When applied to the specific tensor $\psi(x)$, the term containing the gauge field W_μ inherently mixes the N Dirac fermions contained within ψ , and thus W_μ is represented by $N \times N$ matrices (specifically, one matrix for any given value of the spacetime index μ). It performs infinitesimal group transformations and defines a specific sort of parallel transport. It can be conveniently expanded in terms of the group generators as follows:

$$W_\mu(x) = -iW_\mu^a(x)T^a.\tag{1.2.4}$$

This relation explicitly defines the anti-hermitian component gauge fields $W_\mu^a(x)$, and there are exactly $N^2 - 1$ of them for the gauge group $SU(N)$. From the fundamental physical requirement

of covariance:

$$\begin{aligned}\psi(x) &\rightarrow \psi'(x) = U(x)\psi(x), \\ D_\mu\psi(x) &\rightarrow D'_\mu\psi'(x) = U(x)D_\mu\psi(x),\end{aligned}$$

one systematically obtains the following necessary transformation rule for the gauge potential itself:

$$W_\mu(x) \rightarrow W'_\mu(x) = U(x)W_\mu(x)U^{-1}(x) + U(x)\partial_\mu U^{-1}(x). \quad (1.2.5)$$

Let us explicitly prove this relation. By rigorously requiring that $D'_\mu\psi' = UD_\mu\psi$, one computes:

$$\begin{aligned}D'_\mu\psi' &\equiv (\partial_\mu + W'_\mu)\psi' \\ &= UD_\mu\psi = U\partial_\mu\psi + UW_\mu\psi = U\partial_\mu(U^{-1}U\psi) + UW_\mu U^{-1}U\psi \\ &= (U\partial_\mu U^{-1})\psi' + \partial_\mu\psi' + UW_\mu U^{-1}\psi' \\ &= [\partial_\mu + (UW_\mu U^{-1} + U\partial_\mu U^{-1})]\psi',\end{aligned} \quad (1.2.6)$$

which definitively proves the above transformation rule for W_μ . An important clarifying comment: as ψ transforms specifically in the fundamental representation of $SU(N)$, the generators T^a contained within the W_μ of Eqs.(1.2.4) and (1.2.6) are inherently the generators in the fundamental representation. In general application, one must always use the generators corresponding to the representation of the specific tensors on which the covariant derivative acts.

Covariant derivatives do not commute. This profound geometric fact allows us to systematically define the "curvature" tensor (or physical "field strength") in the following way:

$$[D_\mu, D_\nu]\psi = F_{\mu\nu}\psi, \quad (1.2.7)$$

so that, upon expansion

$$\begin{aligned}[D_\mu, D_\nu]\psi &= (\partial_\mu + W_\mu)(\partial_\nu + W_\nu)\psi - (\partial_\nu + W_\nu)(\partial_\mu + W_\mu)\psi \\ &= (\partial_\mu W_\nu - \partial_\nu W_\mu)\psi + W_\mu\partial_\nu\psi - W_\nu\partial_\mu\psi + W_\mu W_\nu\psi - W_\nu W_\mu\psi \\ &= \dots \\ &= (\partial_\mu W_\nu - \partial_\nu W_\mu + [W_\mu, W_\nu])\psi,\end{aligned}$$

we find:

$$F_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + [W_\mu, W_\nu]. \quad (1.2.8)$$

It is mathematically immediate to check that the resulting object transforms covariantly as:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = U F_{\mu\nu} U^{-1}, \quad (1.2.9)$$

which follows logically from the foundational covariance of (1.2.7), endowing the field with its tensorial properties. This specific transformation rule corresponds identically to the adjoint representation (the representation of total dimension $N^2 - 1$).

1.2.4 | Gauge invariant action

It is now remarkably simple to construct a globally and locally gauge invariant lagrangian starting from (1.2.1): it is enough to formally substitute standard derivatives with gauge covariant derivatives (this general procedure is also classically called "minimal coupling"). One immediately finds:

$$\mathcal{L}_1 = -\bar{\psi}(\gamma^\mu D_\mu + m)\psi \quad (1.2.10)$$

that dynamically depends on the new physical field W_μ contained inherently in the covariant derivative D_μ . Now, one must necessarily introduce proper dynamics for W_μ as well. By considering the absolute simplest gauge and Lorentz invariant lagrangian term containing at most two derivatives, one is logically led to:

$$\mathcal{L}_2 = \frac{1}{2} \text{tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} \quad (1.2.11)$$

where we have explicitly used the generators in the fundamental representation properly normalized by $\text{tr} T^a T^b = \frac{1}{2} \delta^{ab}$. We have chosen here a specific canonical normalization. More generally, one should realistically consider a relative mathematical weight between the different gauge-invariant pieces, so that by formally introducing a physical coupling constant g , one writes the final complete lagrangian as:

$$\mathcal{L} = \frac{1}{2g^2} \text{tr}(F_{\mu\nu} F^{\mu\nu}) - \bar{\psi}(\gamma^\mu D_\mu + m)\psi \quad (1.2.12)$$

which is completely gauge invariant under the following comprehensive transformation rules:

$$\begin{aligned} \psi(x) &\rightarrow \psi'(x) = U(x)\psi(x) \\ \bar{\psi}(x) &\rightarrow \bar{\psi}'(x) = \bar{\psi}(x)U^{-1}(x) \\ W_\mu(x) &\rightarrow W'_\mu(x) = U(x)W_\mu(x)U^{-1}(x) + U(x)\partial_\mu U^{-1}(x) \end{aligned} \quad (1.2.13)$$

with the very last one logically implying:

$$F_{\mu\nu}(x) \rightarrow F'_{\mu\nu}(x) = U(x)F_{\mu\nu}(x)U^{-1}(x). \quad (1.2.14)$$

Infinitesimal transformation rules. Let us rigorously report the *infinitesimal transformations* as well. By defining the matrix $\alpha \equiv -i\alpha_a T^a$ with Lie parameters $\alpha_a \ll 1$, one concisely describes an infinitesimal transformation in the form:

$$U = e^{i\alpha_a T^a} = e^{-\alpha} = 1 - \alpha + O(\alpha^2) \quad (1.2.15)$$

so that for the corresponding infinitesimal variations of the fields ($\delta\psi(x) \equiv \psi'(x) - \psi(x)$) one systematically finds:

$$\begin{aligned} \delta\psi &= -\alpha\psi, \\ \delta\bar{\psi} &= \bar{\psi}\alpha, \\ \delta W_\mu &= \partial_\mu \alpha + [W_\mu, \alpha] = D_\mu \alpha, \\ \delta F_{\mu\nu} &= [F_{\mu\nu}, \alpha], \end{aligned} \quad (1.2.16)$$

where in the last-but-one line, we clearly recognize *the covariant derivative acting specifically on the adjoint representation*. The explicit infinitesimal form of the gauge transformations is fundamentally needed when studying the required gauge-fixing procedure to properly quantize the theory.

Physical interpretation and form of the lagrangian. Now, let us rewrite the total lagrangian by considering a helpful field redefinition defined by the simple rescaling $W_\mu \rightarrow gW_\mu$. It is primarily used to get a canonical kinetic normalization for the gauge field, which practically amounts to pushing the explicit coupling constant directly in front of the interaction vertices. The shift naturally induces a corresponding redefinition of the field strength $F_{\mu\nu} \rightarrow gF_{\mu\nu}$, with the new modified field strength given by:

$$F_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + g[W_\mu, W_\nu]. \quad (1.2.17)$$

In terms of their isolated components,

$$\begin{aligned} W_\mu(x) &= -iW_\mu^a(x)T^a \\ F_{\mu\nu}(x) &= -iF_{\mu\nu}^a(x)T^a \end{aligned} \quad (1.2.18)$$

and

$$F_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + gf^{abc}W_\mu^b W_\nu^c \quad (1.2.19)$$

and the total lagrangian ultimately takes the more explicit physical form:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} - \bar{\psi}[\gamma^\mu(\partial_\mu - igW_\mu^a T^a) + m]\psi, \quad (1.2.20)$$

where the contraction of the group indices a is always implicitly understood as a trace over the group indices. The first term describes the free propagation of the gauge fields W_μ^a along with their cubic and quartic self-interactions, while the second term describes the free propagation of the fermion fields ψ along with their elementary interaction with the gauge fields W_μ^a .

The physical coupling constant. The physical coupling constant now visibly appears directly in front of the fundamental interaction vertices. The infinitesimal gauge transformations, after systematically redefining the parameters $\alpha^a \rightarrow g\alpha^a$, read explicitly:

$$\begin{aligned} \delta\psi(x) &= ig\alpha^a(x)T^a\psi(x) \\ \delta\bar{\psi}(x) &= -ig\alpha^a(x)\bar{\psi}(x)T^a \\ \delta W_\mu^a(x) &= \partial_\mu\alpha^a(x) + gf^{abc}W_\mu^b(x)\alpha^c(x) = D_\mu\alpha^a(x) \end{aligned} \quad (1.2.21)$$

where T^a are, as always, the generators in the fundamental representation.

The very first term in the explicit action(1.2.20) elegantly describes the free space propagation of the fields W_μ^a (the fundamental non-abelian spin 1 particle) along with associated cubic and quartic self-interactions inherently governed by coupling constants g and g^2 , respectively. A non positive-definite Killing metric would drastically result in a term with kinetic energy that is definitively not positive-definite, and this would structurally not be acceptable: it is physically necessary to consider only compact groups to successfully satisfy this stability request. The crucial second term describes the free propagation of the ψ fields (spin 1/2 particles endowed with non-abelian charges, i.e., "color" charges) working intimately together with an elementary interaction with the gauge field W_μ^a possessing a strength dynamically measured by the coupling constant g . It can be effectively treated perturbatively if g is sufficiently small enough. The conceptual "non-abelian" or "color" charge strictly corresponds to the mathematical representation of the gauge group chosen for the matter fields ψ (in our specific case we have cleanly taken the fundamental representation, but logically any other representation could have been validly chosen as well). Thus, we have explicitly seen how the fundamental principle of local gauge invariance has powerfully allowed us to systematically derive all the interaction vertices between diverse fields of spin 1/2 and 1 directly in terms of the single universal coupling constant g .

It is highly useful to analytically examine in more precise detail the algebraic structure of the infinitesimal transformation laws that we have been rigorously deriving. Introducing explicit matrix indices, the infinitesimal transformation of the fermion field $\psi(x)$ from (1.2.21) is formally written as:

$$\delta\psi^i(x) = ig\alpha^a(x)(T_F^a)^i_j\psi^j(x) \quad (1.2.22)$$

with indices $i, j = 1, \dots, N$, and where by $(T_F^a)^i_j$ we now clearly indicate the individual matrix elements of the generators in the fundamental representation ($N \times N$ matrices, exactly one for each value of the group index a). Similarly, the transformation of the Dirac conjugate field (morally acting as the complex conjugate field³) is rigorously written as:

$$\delta \bar{\psi}_i(x) = -ig\alpha^a(x)(T_F^{a*})^j_i \bar{\psi}_j(x) = ig\alpha^a(x)(T_F^a)^j_i \bar{\psi}_j(x) \quad (1.2.23)$$

where $T_F^a = -T_F^{a*}$ strictly are the generators residing in the complex conjugate of the fundamental representation T_F^a . Also, we may meticulously check that the non-abelian field strength $F_{\mu\nu}^a$ correctly transforms strictly in the adjoint representation:

$$\delta F_{\mu\nu}^a = gf^{abc}F_{\mu\nu}^b\alpha^c = ig\alpha^c(T_{\text{Adj}}^c)^{ab}F_{\mu\nu}^b \quad (1.2.24)$$

precisely as the generators in the defined adjoint representation are rigorously given by:

$$(T_{\text{Adj}}^c)^{ab} = -if^{abc}. \quad (1.2.25)$$

Let us also verify explicitly and algebraically that the established transformation law of W_μ^a can be completely expressed exactly in terms of the covariant derivative acting dynamically on a tensor situated in the adjoint representation:

$$\delta W_\mu^a = \partial_\mu \alpha^a + gf^{abc}W_\mu^b\alpha^c = \partial_\mu \alpha^a - igW_\mu^b(T_{\text{Adj}}^b)^{ac}\alpha^c = D_\mu \alpha^a. \quad (1.2.26)$$

Note critically also that the isolated non-derivative part of this specific transformation can be formally written equivalently as:

$$\delta W_\mu^a = ig\alpha^c(T_{\text{Adj}}^c)^{ab}W_\mu^b + \dots \quad (1.2.27)$$

which prominently highlights the underlying tensorial character of this specific part of the transformation, perfectly matching (1.2.24).

Finally, let us formally recall that the continuous Jacobi identities for completely arbitrary operators, once carefully applied directly to the covariant derivative operator $D_\mu = \partial_\mu + gW_\mu$

$$[D_\mu, [D_\nu, D_\lambda]] + [D_\nu, [D_\lambda, D_\mu]] + [D_\lambda, [D_\mu, D_\nu]] = 0, \quad (1.2.28)$$

give geometric rise directly to the so-called mathematical Bianchi identities strictly governing the field strength $F_{\mu\nu}$:

$$D_\mu F_{\nu\lambda} + D_\nu F_{\lambda\mu} + D_\lambda F_{\mu\nu} = 0, \quad (1.2.29)$$

where the relevant covariant derivative strictly evaluated in the adjoint representation is analytically written as:

$$D_\mu F_{\nu\lambda} = \partial_\mu F_{\nu\lambda} + g[W_\mu, F_{\nu\lambda}] \quad (1.2.30)$$

that, when broken down in explicit components, structurally takes the complete form:

$$D_\mu F_{\nu\lambda}^a = \partial_\mu F_{\nu\lambda}^a + gf^{abc}W_\mu^b F_{\nu\lambda}^c. \quad (1.2.31)$$

³We logically recall here that upper dotted indices are mathematically equivalent to lower undotted indices for the defining $SU(N)$ group.

1.2.5 | The action of quantum chromodynamics (QCD)

The Gauge Group and Gluons

The fundamental action of quantum chromodynamics (QCD) is built upon the non-abelian gauge group $SU(3)$. In addition to the gluons—the eight spin-1 particles associated with the gauge field W_μ^a , which inherently possess an index in the adjoint representation and thus belong to the **8** representation of $SU(3)$ —the total lagrangian contains six distinct fermion fields ψ_f corresponding to the six known flavors of quarks, denoted by $f = (u, d, c, s, t, b)$. These correspond to the up, down, charm, strange, top, and bottom quarks, respectively.

Each individual quark flavor is degenerate in the sense that it transforms in the fundamental **3** representation of the $SU(3)$ gauge group: the quark is physically said to be "colored" (typically denoted with colors red, green, and blue, following the usual convention). The absence of color geometrically indicates a scalar under the gauge group, much like the total lagrangian itself (a scalar quantity rigorously corresponds to the trivial representation **1** of $SU(3)$).

Naturally, the corresponding antiparticles, the antiquarks $(\bar{u}, \bar{d}, \bar{c}, \bar{s}, \bar{t}, \bar{b})$, transform in the complex conjugate representation, the **$\bar{3}$** of $SU(3)$, which mathematically corresponds to the representation of $\bar{\psi}_f$, the Dirac conjugate of ψ_f ⁴.

Gell-Mann Matrices and the QCD Lagrangian

The eight infinitesimal generators of $SU(3)$ evaluated in the fundamental representation are explicitly given by the Gell-Mann matrices λ^a , which systematically generalize the well-known Pauli matrices σ^i of $SU(2)$ to the higher-dimensional $SU(3)$ case:

$$T^a = \frac{\lambda^a}{2} \quad a = 1, \dots, 8 \quad (1.2.32)$$

where these matrices are defined as:

$$\begin{aligned} \lambda^1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda^3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \lambda^4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda^5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \\ \lambda^6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & \lambda^7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda^8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \end{aligned} \quad (1.2.33)$$

These specific generators are mathematically normalized so that they satisfy $\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$.

An arbitrary element of the $SU(3)$ group residing in the fundamental representation is therefore formally given by 3×3 matrices taking the form $U = \exp(i\alpha_a \frac{\lambda^a}{2})$. By explicitly calculating the Lie algebra relations $[T^a, T^b] = if^{abc} T^c$, one systematically finds the explicit values of the

⁴Given the defining matrices in the fundamental representation $U = e^{i\alpha^a T_F^a}$, the complex conjugate matrices also elegantly furnish a valid representation $U^* = e^{-i\alpha^a T_F^{a*}} = e^{i\alpha^a (-T_F^{a*})} \equiv e^{i\alpha^a T_{\bar{F}}^a}$ so that $T_{\bar{F}}^a = -T_F^{a*} = -T_F^{aT}$, where the last relation follows directly because the established generators are hermitian.

structure constants f^{abc} belonging to the $SU(3)$ group. They are totally antisymmetric under any permutation of indices and are given by:

$$\begin{aligned} f^{123} &= 1 \\ f^{147} &= -f^{156} = f^{246} = f^{257} = f^{345} = -f^{367} = \frac{1}{2} \\ f^{458} &= f^{678} = \frac{\sqrt{3}}{2} \end{aligned}$$

while all other f^{abc} strictly not related to these by permuting the indices are identically zero. As an instructive exercise, one may try to explicitly verify some of these numbers.

The complete **QCD lagrangian** is therefore constructed as:

$$\begin{aligned} \mathcal{L}_{QCD} &= -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} - \sum_{f=1}^6 \bar{\psi}_f (\gamma^\mu D_\mu + m_f) \psi_f \\ &= -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} - \sum_{f=1}^6 \bar{\psi}_f (\gamma^\mu \partial_\mu + m_f) \psi_f + i\frac{g_s}{2} W_\mu^a \sum_{f=1}^6 \bar{\psi}_f \gamma^\mu \lambda^a \psi_f \\ &= \text{diagram 1} + \text{diagram 2} + \text{diagram 3} + \sum_{f=1}^6 \text{diagram 4} + \sum_{f=1}^6 \text{diagram 5}, \end{aligned} \quad (1.2.34)$$

TODO: think about removing or not labels since the legend is already there

where the strong coupling constant is conventionally denoted by g_s , and the sum index $f \in (1, 2, \dots, 6) = (u, d, c, s, t, b)$ specifically indicates the distinct flavor of the quark. Different flavors of quarks physically possess different masses m_f . Note carefully that to obtain the valid mathematical propagator of the gauge field from the very first term, as indicated conceptually in the corresponding figure, one must first rigorously implement a formal gauge-fixing procedure.

Flavor Symmetries and Symmetries of the Action

The QCD lagrangian dynamically possesses additional rigid global symmetries. A well-known rigid symmetry is the $U(1)$ symmetry which uniformly rotates all fields of the quarks by the exact same overall phase ($\psi_f \rightarrow \psi'_f = e^{i\alpha/3} \psi_f$ for all f , where the factor of $\frac{1}{3}$ is strictly conventional): the associated conserved physical charge is the *baryon number* B . It is a robust symmetry that is faithfully preserved also by the other fundamental interactions evaluated at the perturbative level (nonperturbative electroweak transitions dynamically violate $B+L$ while strictly preserving $B-L$ with L denoting the lepton number).

Other independent $U(1)$ symmetries rotate the various fermionic fields entirely separately. They logically give rise to precise conservation laws of the respective *fermion numbers* (e.g. strangeness $\mathcal{S}$, charm $\mathcal{C}$, etc.). These distinct flavor symmetries are exactly preserved for QCD (and QED), but the weak force structurally violates them. There are exactly six $U(1)$ independent conserved charges, one for each specific quark flavor, and the global baryon number B is mathematically a particular linear combination of these six independent initial charges. Also, the physical *electric charge* Q is a specific linear combination of them, and it is precisely the one which is eventually locally gauged to correctly obtain the electromagnetic couplings.

A comprehensive summary of these $U(1)$ symmetries is explicitly given in the following table, which rigorously reports the various $U(1)$ charges with a widely accepted standard normalization:

Quarks	$\mathcal{U}$	$\mathcal{D}$	$\mathcal{C}$	$\mathcal{S}$	$\mathcal{T}$	$\mathcal{B}$	B	Q
u	1	0	0	0	0	0	$\frac{1}{3}$	$\frac{2}{3}$
d	0	-1	0	0	0	0	$\frac{1}{3}$	$-\frac{1}{3}$
c	0	0	1	0	0	0	$\frac{1}{3}$	$\frac{2}{3}$
s	0	0	0	-1	0	0	$\frac{1}{3}$	$-\frac{1}{3}$
t	0	0	0	0	1	0	$\frac{1}{3}$	$\frac{2}{3}$
b	0	0	0	0	0	-1	$\frac{1}{3}$	$-\frac{1}{3}$

Note critically that we have indicated the baryon number explicitly by B , and the bottom (or beauty) quantum number by $\mathcal{B}$. For each continuous symmetry, each quark flavor formally transforms with the specific charge indicated in the table. For example, for the physical electric charge Q we have the transformation:

$$\psi_f \rightarrow \psi'_f = e^{i\alpha Q_f} \psi_f. \quad (1.2.35)$$

By carefully looking at the generated table, one immediately recognizes the following algebraic relations:

$$\begin{aligned} B &= \frac{1}{3}(\mathcal{U} + \mathcal{C} + \mathcal{T}) - \frac{1}{3}(\mathcal{D} + \mathcal{S} + \mathcal{B}) \\ Q &= \frac{2}{3}(\mathcal{U} + \mathcal{C} + \mathcal{T}) + \frac{1}{3}(\mathcal{D} + \mathcal{S} + \mathcal{B}). \end{aligned} \quad (1.2.36)$$

Isospin and the Eightfold Way

There are also highly useful approximate symmetries characterizing the QCD lagrangian. In the theoretical limit in which some of the quark masses are mathematically taken to be strictly identical, there emerges a rigid additional non-abelian symmetry. For example, assuming identically matched masses for the up and down quarks, $m_u = m_d$, one can continuously mix the specific fields ψ_u and ψ_d with a $SU(2)$ matrix to effectively redefine what is physically meant by “up” and “down”:

$$\begin{pmatrix} \psi_u \\ \psi_d \end{pmatrix} \rightarrow \begin{pmatrix} \psi'_u \\ \psi'_d \end{pmatrix} = U \begin{pmatrix} \psi_u \\ \psi_d \end{pmatrix} \quad U \in SU(2). \quad (1.2.37)$$

This rigid $SU(2)$ symmetry directly corresponds to the **strong isospin** $\vec{I}$, successfully used to properly group hadrons into distinct families or multiplets. From physical phenomenology, one knows that the strong force tightly binds states of quarks together and clearly shows the profound phenomenon of color confinement: physically bound states of quarks are always color singlets and directly correspond to the observable mesons and baryons. Key examples of these families are: (i) the isospin doublet of the core nucleons (proton and neutron) structurally composed of three confined up and down quarks; (ii) the triplet of π mesons, the pions $\pi^\pm$ and π^0 , fundamentally composed of a quark and an antiquark entirely of the up and down types.

Considering structurally identical masses for the quarks up, down, and strange, $m_u = m_d = m_s$, one logically finds an *even larger symmetry group*, the $SU(3)$ flavor group, that continuously mixes

the three flavors up, down, and strange:

$$\begin{pmatrix} \psi_u \\ \psi_d \\ \psi_s \end{pmatrix} \rightarrow \begin{pmatrix} \psi'_u \\ \psi'_d \\ \psi'_s \end{pmatrix} = U \begin{pmatrix} \psi_u \\ \psi_d \\ \psi_s \end{pmatrix} \quad U \in SU(3). \quad (1.2.38)$$

This $SU(3)$ flavor group is precisely the one that is heavily used in the static quark model (the famous “**eightfold way**” of Gell-Mann) to neatly take care of the numerous similarities observationally observed between the various hadrons. It should strictly not be confused with the foundational color group, which is dynamically also an $SU(3)$ group. As already emphatically said, color is expected to tightly confine inside the hadrons and thus leaves only composite colorless states physically observable. Key examples of observable multiplets of hadronic particles successfully described by the $SU(3)$ flavor group are:

- the meson octet $(\pi^\pm, \pi^0, K^\pm, K^0, \bar{K}^0, \eta_8)$,
- the baryon octet $(p, n, \Sigma^\pm, \Sigma^0, \Xi^0, \Xi^-, \Lambda)$,
- the baryon decuplet $(\Delta^-, \Delta^0, \Delta^+, \Delta^{++}, \Sigma^{*\pm}, \Sigma^{*0}, \Xi^{*0}, \Xi^{*-}, \Omega^-)$.

The physical existence of these structured families is entirely compatible with formal group theory: the **8** and the **10** are well-known representations of $SU(3)$. Let us logically consider the mesons in more precise detail. They phenomenologically consist of a quark-antiquark pair $(q\bar{q})$. The quarks q transform dynamically in the **3** of $SU(3)$, with $\mathbf{3} \sim (u, d, s)$, while the antiquarks $\bar{q}$ transform mathematically in the $\bar{\mathbf{3}}$ of $SU(3)$, with $\bar{\mathbf{3}} \sim (\bar{u}, \bar{d}, \bar{s})$. From this, it logically follows that possible physically bound states $(q\bar{q})$ must transform in the tensor product space:

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$$

and therefore both singlet and octets could, in theoretical principle, exist physically for the mesons.

Baryons are tightly bound states of three quarks (qqq) , and since $SU(3)$ group theory mathematically tells us that

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = (\mathbf{6} \oplus \bar{\mathbf{3}}) \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$$

we intuitively understand that octets and decuplets are formally allowed structural possibilities for multiplets of baryons.

It is also theoretically useful to consider a noticeably simpler version of QCD strictly containing only three massless flavours of fermions. This can often provide a highly good approximation, since the actual physical masses of the three lightest quarks can often be reasonably neglected in many high-energy regimes. In this idealized limit, the full global internal symmetry is:

$$U(3)_L \times U(3)_R = U(3)_V \times U(3)_A = U(1)_V \times SU(3)_V \times U(1)_A \times SU(3)_A. \quad (1.2.39)$$

The axial symmetry $SU(3)_A$ is dynamically spontaneously broken, logically giving rise to eight purely massless Goldstone bosons, which are phenomenologically identified with the meson octet explicitly described above. By contrast, the $U(1)_A$ symmetry is fundamentally broken by the quantum axial anomaly and therefore does not survive as a true symmetry at the fully quantum level. Consequently, strictly no corresponding physical Goldstone boson is expected.

1.3 | Quantization of Gauge Theories

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TODO: Add introduction to the quantization of gauge theories.

1.3.1 | Faddeev Popov Quantization

The quantization of gauge theories necessitates special mathematical considerations to systematically construct a well-defined Quantum Field Theory (QFT) and its corresponding perturbative expansion. The fundamental problem becomes immediately clear within the path integral approach, where a naive path integral quantization yields a severely diverging result. Let us consider the illustrative case of the abelian $U(1)$ theory, which describes free electromagnetism.

The classical action and the naive path integral are given by:

$$\begin{aligned} S[A] &= \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right), \\ Z &= \int \mathcal{D}A e^{iS[A]} \rightarrow \infty, \end{aligned} \quad (1.3.1)$$

where the electromagnetic field strength tensor is defined as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Here, the path integral inherently diverges because one is redundantly summing over an infinite number of gauge-equivalent field configurations. Specifically, configurations related by the local gauge transformation

$$A_\mu(x) \rightarrow A_\mu^g(x) = A_\mu(x) + ig(x)\partial_\mu g^{-1}(x), \quad g(x) = e^{i\alpha(x)} \in U(1), \quad (1.3.2)$$

possess the exact same value of the action, $S[A^g] = S[A]$. Consequently, the entire field space dynamically decomposes into inequivalent continuous **gauge orbits**, as visualized conceptually in the figure below.

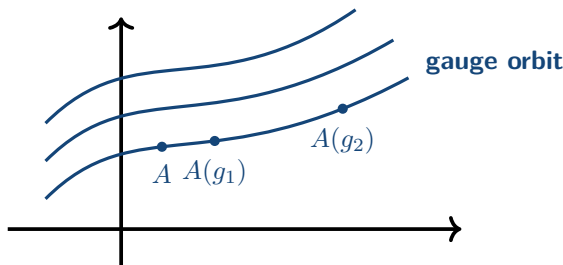


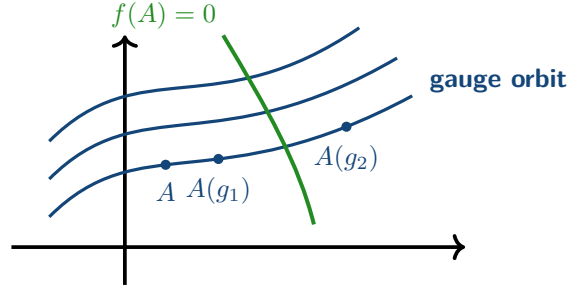
Figure 1.1: Gauge orbits in field space. The field configurations A , $A(g_1)$, and $A(g_2)$ lie on the same gauge orbit, meaning they are physically equivalent and connected by a gauge transformation $g(x) \in U(1)$.

One would ideally like to formally define the path integral in such a way as to obtain a finite and explicitly gauge-invariant result by factoring out this redundant infinite volume:

$$Z = \int \frac{\mathcal{D}A}{\text{Vol}(\text{Gauge})} e^{iS[A]} \sim \text{finite}, \quad (1.3.3)$$

where the formal quantity “Vol(Gauge)” indicates the infinite volume of the gauge group. This conceptual definition can be implemented concretely using a *gauge-fixing function à la Faddeev-Popov*, where unphysical “ghost” fields are elegantly introduced to exponentiate a crucial measure factor. Ideally, the chosen gauge-fixing function should pick exactly one single representative from each gauge orbit, as sketched in figure 1.2. This idealized situation precisely corresponds to the choices of the so-called “**unitary gauges**”. However, for practical perturbative calculations, it is often enough to merely reduce the number of representatives for each gauge orbit, which naturally happens in the standard choices of **covariant gauges**.

Figure 1.2: Gauge fixing by the condition $f(A) = 0$. The gauge-fixing function intersects the field space such that it picks ideally just one representative from each gauge orbit.



As we shall explore, the gauge-fixed action thus obtained using covariant gauges exhibits a powerful residual global symmetry, called BRST symmetry. This symmetry is a direct remnant of the original gauge symmetry and is deeply instrumental in rigorously proving many properties of quantum gauge theories. In particular, it can be utilized to formally prove the independence of physical, observable quantities from the arbitrary choice of gauge-fixing function. It also forms the foundational basis of the BRST lagrangian quantization method, which systematically generalizes the Faddeev-Popov method to more complex, general cases. An even more powerful advanced method utilizes additional fields (the so-called antifields) and is widely known as the Batalin-Vilkovisky method.

The Faddeev-Popov Trick

Let us start by explicitly describing the Faddeev-Popov method in the simpler free abelian gauge theory case. The clever trick is to use a gauge-fixing condition, as depicted in figure 1.2, and insert the identity (written in a suitable integral way) directly into the path integral to effectively extract the volume of the gauge group. Consider the elementary 1-dimensional identity written as:

$$1 = \int df \delta(f) = \int \frac{\partial f(y)}{\partial y} dy \delta(f(y)) = \int dy \delta(f(y)) \frac{\partial f(y)}{\partial y}$$

which smoothly generalizes to n -dimensions where f becomes a vector f^i with n components:

$$1 = \int d^n f \delta^{(n)}(f) = \int d^n y \delta^{(n)}(f(y)) \det \left(\frac{\partial f^i(y)}{\partial y^j} \right).$$

We can formally generalize this finite-dimensional expression further to functional integrals over continuous fields:

$$1 = \int \mathcal{D}g \delta(f(A^g(x))) \text{Det} \left(\frac{\delta f(A^g(x))}{\delta g(y)} \right) \quad (1.3.4)$$

where $\mathcal{D}g$ is a suitable gauge-invariant measure that also fundamentally produces the volume of the gauge group, $\int \mathcal{D}g = \text{Vol}(\text{Gauge})$. The notation “Det” strictly indicates a functional determinant.

The delta function (which is actually a delta *functional*) $\delta(f(x))$ mathematically means that the whole function $f(x)$ is strictly set to vanish.

Then, let us compute formally the path integral defined in (1.3.3) by systematically inserting the functional identity from (1.3.4):

$$\begin{aligned}
Z &= \int \frac{\mathcal{D}A}{\text{Vol}(\text{Gauge})} e^{iS[A]} \\
&= \int \frac{\mathcal{D}A}{\text{Vol}(\text{Gauge})} \int \mathcal{D}g \delta(f(A^g(x))) \text{Det} \left(\frac{\delta f(A^g(x))}{\delta g(y)} \right) e^{iS[A]} \\
&= \int \frac{\mathcal{D}A^g \mathcal{D}g}{\text{Vol}(\text{Gauge})} \delta(f(A^g(x))) \text{Det} \left(\frac{\delta f(A^g(x))}{\delta g(y)} \right) e^{iS[A^g]} \\
&= \int \frac{\mathcal{D}g}{\text{Vol}(\text{Gauge})} \int \mathcal{D}A \delta(f(A(x))) \text{Det} \left(\frac{\delta f(A^g(x))}{\delta g(y)} \right) \Big|_{g=1} e^{iS[A]} \\
&= \int \mathcal{D}A \delta(f(A(x))) \text{Det} \left(\frac{\delta f(A^g(x))}{\delta g(y)} \right) \Big|_{g=1} e^{iS[A]}
\end{aligned}$$

which is precisely the gauge-fixed path integral we were actively looking for. In these algebraic manipulations, we have first inserted the identity (1.3.4), and then utilized the fundamental fact that the action and the measure are both completely gauge invariant, namely $S[A^g] = S[A]$ and $\mathcal{D}A^g = \mathcal{D}A$. This is certainly mathematically true for the classical action, but it is an implicit assumption for the measure (more concretely, it is intimately related to the regularization methods used to make strict sense of the diverging Feynman diagrams characterizing the perturbative expansion⁵). Then, we have formally changed integration variables from A^g to A , so that nothing in the integrand depends on the gauge parameter $g(x)$ anymore. Consequently, the integration on $\mathcal{D}g$ can be cleanly factorized out to exactly cancel the infinite gauge volume $\text{Vol}(\text{Gauge})$ ⁶.

The final formula is theoretically correct but can be elegantly written in a significantly more useful form by performing two essential steps:

- (i) introducing **ghost fields**, i.e., anticommuting fields $c(x)$ and $\bar{c}(x)$ to exponentiate the functional determinant, broadly known as the Faddeev-Popov determinant,
- (ii) modifying the gauge-fixing function to effectively get rid of the rigid delta functional from the path integral: instead of strictly setting $f(A(x)) = 0$, one may equivalently set it to $f(A(x)) = h(x)$ (where $h(x)$ is an arbitrary continuous function) and then functionally average over the diverse functions $h(x)$ with a gaussian weight $e^{-\frac{1}{2\xi} \int h^2}$.

The rigorous physical rationale behind this averaging procedure is that inherently different gauge-fixing functions must ultimately give the exact same gauge-invariant result. In particular, nothing physical should depend on the specific numerical value of the gauge parameter ξ .

Thus, implementing these exponentiation steps, one initially finds:

$$\begin{aligned}
Z &= \int \mathcal{D}A \mathcal{D}c \mathcal{D}\bar{c} \mathcal{D}h \delta(f(A(x)) - h(x)) \\
&\quad \times \exp i \left(S[A] + \int d^4x d^4y \bar{c}(x) \frac{\delta f(A^g(x))}{\delta g(y)} \Big|_{g=1} c(y) - \frac{1}{2\xi} \int d^4x h^2(x) \right),
\end{aligned}$$

⁵In chiral theories where the chosen regularization might fail to actively maintain gauge invariance, one physically speaks of **anomalies**. The entire quantization procedure breaks down if the potential anomalies in the gauge symmetry are not exactly canceled.

⁶The formal mathematical identification $\delta g \sim g^{-1} dg$ yields a strictly group-invariant measure, the so-called Haar measure.

which is drastically simplified by path integrating over the auxiliary function $h(x)$ to cleanly eliminate the delta functional, allowing us to find in the exponent the complete gauge-fixed total action S_{tot} :

$$\begin{aligned} Z &= \int \mathcal{D}A \mathcal{D}c \mathcal{D}\bar{c} \exp i \left(S[A] + \int d^4x d^4y \bar{c}(x) \frac{\delta f(A^g(x))}{\delta g(y)} \Big|_{g=1} c(y) - \frac{1}{2\xi} \int d^4x f^2(A(x)) \right) \\ &= \int \mathcal{D}A \mathcal{D}c \mathcal{D}\bar{c} e^{i S_{\text{tot}}[A, c, \bar{c}]}, \end{aligned} \quad (1.3.5)$$

where the action appearing in the exponent has to be interpreted as the *complete gauge-fixed action*.

Example (The Lorenz Gauge). To exemplify the abstract construction derived above, let us explicitly choose a specific gauge-fixing function:

$$f(A) = \partial^\mu A_\mu, \quad (1.3.6)$$

which corresponds directly to the strict *Lorenz gauge* for the initial path integral derivation, and to a weighted Lorenz gauge for the path integral in (1.3.5) (also frequently called the R_ξ gauge). Under an infinitesimal gauge variation $\delta A_\mu = \partial_\mu \alpha$, the gauge-fixing function varies algebraically as:

$$\delta f(A) = \partial^\mu \delta A_\mu = \partial^\mu \partial_\mu \alpha,$$

and one is mathematically led to naturally interpret the functional derivative operator as:

$$\frac{\delta f(A^g(x))}{\delta g(y)} \Big|_{g=1} \sim \frac{\delta f(A(x))}{\delta \alpha(y)} = \partial^\mu \partial_\mu \delta^4(x - y),$$

whose functional determinant (the Faddeev-Popov determinant) can be faithfully reproduced by path integrating over the anti-commuting scalar ghost fields $c(x)$ and $\bar{c}(x)$. Thus, the complete gauge-fixed action reads:

$$S_{\text{tot}}[A, c, \bar{c}] = S[A] + \int d^4x \left(-\partial^\mu \bar{c} \partial_\mu c - \frac{1}{2\xi} (\partial^\mu A_\mu)^2 \right), \quad (1.3.7)$$

which is also called the *quantum action* of the theory.

The corresponding total gauge-fixed lagrangian density is given by:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \partial^\mu \bar{c} \partial_\mu c - \frac{1}{2\xi} (\partial^\mu A_\mu)^2, \quad (1.3.8)$$

which, up to total derivative terms that are usually dropped in the action since they do not affect equations of motion, may be symmetrically written as:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{2} \partial^\mu A^\nu \partial_\mu A_\nu + \frac{1}{2} \left(1 - \frac{1}{\xi} \right) (\partial^\mu A_\mu)^2 - \partial^\mu \bar{c} \partial_\mu c. \quad (1.3.9)$$

In particular, the highly convenient Feynman gauge is obtained by setting $\xi = 1$ and yields the remarkably simple action:

$$\mathcal{L}_{\text{tot}} = -\frac{1}{2} \partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu \bar{c} \partial_\mu c. \quad (1.3.10)$$

Because the path integral is now perfectly well-defined, one can safely add external sources and compute the fundamental propagators of the theory. In the Feynman gauge, they read in momentum space:

$$\begin{aligned} \langle A_\mu(x) A_\nu(y) \rangle &= \int \frac{d^4p}{(2\pi)^4} e^{ip(x-y)} \frac{-i\eta_{\mu\nu}}{p^2 - i\epsilon} \\ \langle c(x) \bar{c}(y) \rangle &= \int \frac{d^4p}{(2\pi)^4} e^{ip(x-y)} \frac{-i}{p^2 - i\epsilon}. \end{aligned} \quad (1.3.11)$$

Exercise: Find the exact form of the propagators in a generic R_ξ gauge.

At this advanced stage, one can also add matter fermions to obtain the complete QED action and critically note that the ghost fields can be safely integrated out and completely eliminated from the abelian theory. This is fundamentally because they contribute at most to an overall constant normalization factor that completely disappears when evaluating physically normalized correlation functions. This will decidedly not be the case for non-abelian gauge theories (and also for QED formulated in curved spacetime), but before mathematically addressing the more complex non-abelian case, let us discover and rigorously study a rigid symmetry that is robustly present in the gauge-fixed action and which has far-reaching physical implications, the BRST symmetry.

1.3.2 | BRST Symmetry

Once the gauge symmetry has been fixed, the resulting gauge-fixed action S_{tot} is no longer gauge invariant. This is a necessary consequence, as the gauge-fixing term is explicitly introduced to break this redundancy and define a nonsingular path integral. However, a remnant of the original gauge symmetry survives in the form of BRST (Becchi-Rouet-Stora-Tyutin) symmetry, which is a rigid, global symmetry that depends on a constant, anticommuting (Grassmann-valued) parameter Λ .

Let us consider the total lagrangian in the Feynman gauge:

$$\mathcal{L}_{tot} = -\frac{1}{2}\partial^\mu A^\nu \partial_\mu A_\nu - \partial^\mu \bar{c} \partial_\mu c \quad (1.3.12)$$

Under the BRST symmetry, the fields transform according to the following rules:

$$\begin{cases} \delta_B A_\mu(x) = \Lambda \partial_\mu c(x) \\ \delta_B c(x) = 0 \\ \delta_B \bar{c}(x) = \Lambda \partial^\mu A_\mu(x) \end{cases} \quad (1.3.13)$$

These transformations ensure that the variation of the total lagrangian vanishes up to total derivatives, meaning $\delta_B \mathcal{L}_{tot} = 0$. It is important to recall that the transformation parameter Λ , as well as the ghost fields $c(x)$ and $\bar{c}(x)$, are all Grassmann-valued quantities, which means they mutually anticommute.

A crucial, defining property of this symmetry is that it is nilpotent. Specifically, it is a Lie symmetry featuring a Grassmann odd parameter that satisfies the commutator relation:

$$[\delta_B(\Lambda_1), \delta_B(\Lambda_2)] = 0 \quad (1.3.14)$$

where one evaluates the variation using two different values of the constant Grassmann parameter, Λ_1 and Λ_2 .

The nilpotency property becomes much more evident if one defines the Slavnov operator s by formally factorizing out the constant parameter Λ :

$$\delta_B(\Lambda) = \Lambda s. \quad (1.3.15)$$

The Slavnov operator s acts as a nilpotent graded variation. In this context, "graded" indicates that the operator anticommutes with Grassmann odd fields, and "nilpotency" explicitly means that it satisfies the condition:

$$s^2 = 0 \quad (1.3.16)$$

Operating twice with s thus gives a vanishing result on any arbitrary field. Explicitly, extracting s from Eq. (1.3.13), we get:

$$\begin{cases} sA_\mu(x) = \partial_\mu c(x) \\ sc(x) = 0 \\ s\bar{c}(x) = \partial^\mu A_\mu(x) \end{cases} \quad (1.3.17)$$

Applying the operator a second time confirms that it indeed gives zero:

$$s^2 A_\mu(x) = s^2 c(x) = 0 \quad \text{and} \quad s^2 \bar{c}(x) = \partial^\mu \partial_\mu c(x) = 0 \quad (1.3.18)$$

Notice carefully that the classical ghost equations of motion ($\partial^\mu \partial_\mu c = 0$) have been used in the last equation to ensure the result vanishes. Because one strictly needs to use the equations of motion to satisfy $s^2 = 0$, these particular BRST symmetry rules are said to be nilpotent *on-shell*.

As for any rigid continuous symmetry, Noether's theorem guarantees there is an associated conserved current J_B^μ . This current can be found as usual by applying the Noether trick, which involves artificially extending the global parameter to a local one, $\Lambda \rightarrow \Lambda(x)$, in Eq. (1.3.13) and computing the variation of the action:

$$\delta_B S_{tot} = \int d^4x (\partial_\mu \Lambda) J_B^\mu, \quad J_B^\mu = -F^{\mu\nu} \partial_\nu c - \partial^\nu A_\nu \partial^\mu c \quad (1.3.19)$$

The associated conserved BRST charge Q_B is simply the spatial integral of the zero-component of this current:

$$Q_B = \int d^3x J_B^0 = \int d^3x (\partial_\nu A^\nu \dot{c} - F^{0i} \partial_i c). \quad (1.3.20)$$

Upon canonical quantization, this classical charge mathematically becomes a nilpotent quantum operator: $Q_B \rightarrow \hat{Q}_B$, satisfying $\hat{Q}_B^2 = 0$.

There is an elegant theoretical way of adding auxiliary fields to the lagrangian to make the BRST symmetry nilpotent *off-shell* (i.e., without requiring the equations of motion). To see how this remarkable property emerges, let us consider a Fourier path integral representation⁷ of the delta functional utilized during the gauge-fixing procedure:

$$\delta(f(x)) = \int \mathcal{D}B e^{-i \int d^4x B(x) f(x)}, \quad (1.3.21)$$

We then substitute this functional representation directly into the gauge-fixed path integral. Subsequently, we path-integrate over the arbitrary gauge-weighting function h by standard Gaussian integration. The relevant part of the measure manipulations is given by:

$$\begin{aligned} & \int \mathcal{D}h \delta(f(A(x)) - h(x)) e^{-i \int d^4x \frac{1}{2\xi} h^2(x)} \\ &= \int \mathcal{D}h \mathcal{D}B e^{-i \int d^4x (B(x)[f(A(x)) - h(x)] + \frac{1}{2\xi} h^2(x))} \\ &= \int \mathcal{D}B e^{i \int d^4x (-B(x)f(A(x)) + \frac{\xi}{2} B^2(x))}. \end{aligned} \quad (1.3.22)$$

From these algebraic manipulations, one naturally finds a new total lagrangian describing the extended set of fields $(A_\mu, c, \bar{c}, B)$. Given that we have previously taken the covariant gauge-fixing function to be $f(A) = \partial^\mu A_\mu$, the lagrangian takes the updated form:

$$\mathcal{L}_{tot} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \partial^\mu \bar{c} \partial_\mu c - B \partial^\mu A_\mu + \frac{\xi}{2} B^2. \quad (1.3.23)$$

⁷This functional representation directly extends the usual one-dimensional formula for the Dirac delta function $\delta(x) = \int \frac{dk}{2\pi} e^{-ikx}$ to the infinite-dimensional functional case.

With the introduction of the auxiliary field, the BRST symmetry transformations now read:

$$\begin{cases} \delta_B A_\mu = \Lambda \partial_\mu c \\ \delta_B c = 0 \\ \delta_B \bar{c} = \Lambda B \\ \delta_B B = 0 \end{cases} \quad (1.3.24)$$

This expanded set of transformations is manifestly nilpotent off-shell. Let us systematically recall the quantum statistics of the fields involved: the fields A_μ and B are commuting (bosonic), while the ghost fields c and $\bar{c}$ are anticommuting (fermionic). The field $c(x)$ is explicitly called the *ghost* and conceptually corresponds to the local gauge parameter $\alpha(x)$, which in a profound sense is turned into a full dynamical field, but endowed with the opposite Grassmann character ($\alpha(x) \rightarrow \Lambda c(x)$). The field $\bar{c}$ is analogously called the *antighost*, while the newly introduced field B is the *auxiliary field* (also widely known in literature as the Nakanishi-Lautrup field).

Of course, since B does not contain any derivative kinetic terms, one can easily eliminate it by using its purely algebraic equations of motion:

$$B = \frac{1}{\xi} \partial^\mu A_\mu \quad (1.3.25)$$

Substituting this back into the action, or equivalently, path-integrating over the B field in the functional integral, exactly reproduces the original lagrangian given in Eq. (1.3.12) (up to the ξ dependence). This explicitly checks that the new lagrangian provides a physically equivalent description of the exact same theory. However, eliminating the auxiliary field means that the BRST transformations reduce back to:

$$\begin{cases} \delta_B A_\mu = \Lambda \partial_\mu c \\ \delta_B c = 0 \\ \delta_B \bar{c} = \frac{1}{\xi} \partial^\mu A_\mu \Lambda \end{cases} \quad (1.3.26)$$

where strict nilpotency is unfortunately achieved only on-shell once again.

1.3.3 | Non-Abelian Gauge Fields

[...]

1.4 | BRST Quantization

[...]

1.4.1 | Cohomology

[...]

1.4.2 | Application: Gauge-Fixing of Perturbative Quantum Gravity

[...]

1.4.3 | Graviton Propagator in Flat Spacetime

[...]

Exercises

Sheet 1

[...]

Appendices

A | Notation and Conventions

B | Computations and further developments

Notes on group theory

Lie groups and algebras

Given a simple and compact Lie group G , we conventionally indicate its elements using the exponential parametrization $U(\alpha) = \exp(i\alpha_a T^a)$, where the parameters α_a are continuous and T^a are the infinitesimal hermitian generators that fundamentally satisfy the Lie algebra:

$$[T^a, T^b] = if^{ab}_c T^c. \quad (\text{B.1})$$

In general, considering an irreducible unitary representation R of the group G , we automatically obtain a corresponding irreducible representation of its Lie algebra formulated with traceless hermitian matrices T_R^a :

$$[T_R^a, T_R^b] = if^{ab}_c T_R^c. \quad (\text{B.2})$$

The representative matrices T_R^a act linearly on a vector space of dimensions $D(R)$, and thus they are explicitly $D(R) \times D(R)$ square matrices. The integer $D(R)$ is technically called the dimension of the representation. Throughout our discussions, we will mostly consider the special unitary group $SU(N)$, whose main fundamental representations are:

- the fundamental (or defining) representation N , which has a dimension $D(N) = N$
- its complex conjugate representation $\bar{N}$, which also has a dimension $D(\bar{N}) = N$
- the adjoint representation Adj , which possesses a dimension $D(\text{Adj}) = N^2 - 1$.

Given a general representation R populated with generators T_R^a , the generators of its complex conjugate representation $\bar{R}$ are mathematically given by

$$T_{\bar{R}}^a = -T_R^{a*} \quad (\text{B.3})$$

as can be directly seen from taking the complex conjugate of the original representation's group elements:

$$(\exp(i\alpha_a T_R^a))^* = \exp(-i\alpha_a T_R^{a*}) \equiv \exp(i\alpha_a T_{\bar{R}}^a).$$

Killing Metric and Structure Constants

The generators are usually normalized by convention so that in the fundamental representation one explicitly has

$$\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab} \quad (\text{B.4})$$

which structurally fixes the Killing metric, defined generally as $\gamma^{ab} \equiv 2 \text{tr}(T^a T^b)$, to simply be $\gamma^{ab} = \delta^{ab}$. This metric fundamentally defines scalar products within the algebra and is utilized to seamlessly raise or lower the indices that logically label the generators (which strictly correspond to the indices of the adjoint representation). In particular, it is widely used to formally define the structure constants with purely upper indices:

$$f^{abc} = f^{ab}{}_d \delta^{dc} \quad (\text{B.5})$$

(or more generally expressed as $f^{abc} = f^{ab}{}_d \gamma^{dc}$). This fully raised tensor is proven to be totally antisymmetric. The antisymmetry of f^{abc} is immediately obvious on the first two indices, precisely as seen from the definition of the Lie algebra itself. Then, by carefully using Eq. (B.1) and the normalization Eq. (B.4), one can explicitly compute the following intermediate steps:

$$\begin{aligned} \text{tr}([T^a, T^b]T^c) &= i f^{ab}{}_d \text{tr}(T^d T^c) = \frac{i}{2} f^{abc} = \text{tr}(T^a T^b T^c) - \text{tr}(T^b T^a T^c) \\ &= \text{tr}(T^c T^a T^b) - \text{tr}(T^a T^c T^b) = -\text{tr}([T^a, T^c]T^b) = -\frac{i}{2} f^{acb} \end{aligned}$$

so that we find $f^{abc} = -f^{acb}$, which elegantly implies complete antisymmetry across all indices. Note that in the above algebraic manipulations, we have critically used the cyclic property of the trace operation.

The structure constants themselves can be uniquely used to define the adjoint representation ‘Adj’ by setting the matrix elements as

$$(T_{\text{Adj}}^a)^b{}_c = -i f^{ab}{}_c \quad (\text{B.6})$$

since the fundamental Lie algebra relation

$$[T_{\text{Adj}}^a, T_{\text{Adj}}^b] = i f^{ab}{}_c T_{\text{Adj}}^c \quad (\text{B.7})$$

is identically satisfied because of the foundational Jacobi identity.

Representation Indices and Casimir Operators

One mathematically defines the *index* $T(R)$ of an arbitrary representation R by the trace relation

$$\text{tr}(T_R^a T_R^b) = T(R) \delta^{ab}. \quad (\text{B.8})$$

The specific index of the fundamental representation N is usually normalized by convention to $T(N) = \frac{1}{2}$, clearly seen in Eq. (B.4). Once this is fixed, the values of the indices corresponding to other representations are determined unambiguously.

Casimir operators are specialized operators built structurally from the generators which commute cleanly with all the generators of the entire group. In particular, the quadratic Casimir operator that is systematically constructed using the Killing metric

$$C_2 = T^a T^b \gamma_{ab} = T^a T^a \quad (\text{B.9})$$

is precisely such an operator. The proof of its commutation is straightforward and simple:

$$\begin{aligned} [C_2, T^b] &= [T^a T^a, T^b] = T^a [T^a, T^b] + [T^a, T^b] T^a = T^a i f^{abc} T^c + i f^{abc} T^c T^a \\ &= i f^{abc} (T^a T^c + T^c T^a) = 0 \end{aligned}$$

where the final equality holds directly because the structure constants are completely antisymmetric, while the term in parentheses is symmetric⁸. Since C_2 commutes with all the generators, it fundamentally must be proportional to the identity operator in any given irreducible representation (this is a direct consequence of Schur's lemma). This property mathematically defines the specific number $C(R)$, representing the quadratic Casimir evaluated in the irrep R , by

$$T_R^a T_R^a = C(R) \mathbf{1}. \quad (\text{B.10})$$

By cleverly setting $a = b$ in Eq. (B.8) and summing over the indices (i.e., taking the full scalar product utilizing the Killing metric) gives the profound relation

$$C(R)D(R) = T(R)D(\text{Adj}). \quad (\text{B.11})$$

For the absolute simplest representations, one analytically finds

$$D(N) = D(\overline{N}) = N \quad T(N) = T(\overline{N}) = \frac{1}{2} \quad C(N) = C(\overline{N}) = \frac{N^2 - 1}{2N} \quad (\text{B.12})$$

$$D(\text{Adj}) = N^2 - 1 \quad T(\text{Adj}) = N \quad C(\text{Adj}) = N. \quad (\text{B.13})$$

Invariant Tensors and Clebsch-Gordan Coefficients

Finally, it is immensely useful to recall the concept of *invariant tensors*. They are rigorously defined as tensors that remain completely invariant after group transformations are applied. For example, denoting by ψ^i the vectors transforming structurally in the defining representation of $SU(N)$, so that the specific upper index i is dynamically transformed by the defining matrices U_j^i of $SU(N)$, then the familiar Kronecker symbol δ_j^i inherently acts as an invariant tensor:

$$\delta_j^i \rightarrow \delta_j'^i = U_k^i (U^{-1,T})_j^l \delta_l^k = U_k^i (U^*)_j^l \delta_l^k = U_k^i (U^\dagger)_j^k = \delta_j^i.$$

It geometrically tells us that in combining the representation N with $\overline{N}$ there mathematically must appear a scalar

$$N \otimes \overline{N} = \mathbf{1} \otimes \dots \quad (\text{B.14})$$

i.e., one can form the invariant scalar $\psi^i \chi_i$ directly out of ψ^i and χ_i . Similarly, the completely antisymmetric tensor carrying N upper indices, $\epsilon^{i_1 i_2 \dots i_N}$, carefully normalized to $\epsilon^{12 \dots N} = 1$, is a fundamental invariant tensor known also as the Levi-Civita symbol[cite: 15, 16]. Indeed, one verifies algebraically that

$$\epsilon^{i_1 i_2 \dots i_N} \rightarrow \epsilon'^{i_1 i_2 \dots i_N} = U_{j_1}^{i_1} U_{j_2}^{i_2} \dots U_{j_N}^{i_N} \epsilon^{j_1 j_2 \dots j_N} = (\det U) \epsilon^{i_1 i_2 \dots i_N}$$

but since $\det U = 1$ specifically for $SU(N)$, the invariant property cleanly follows. The exact same logic applies for the lower-index version $\epsilon_{i_1 i_2 \dots i_N}$.

Other vital invariant tensors are the generators themselves evaluated in any given nontrivial representation R , which we explicitly write as $(T_R^a)^\alpha_\beta$, where the upper index α is properly associated with the representation R and the lower index β mathematically corresponds to the conjugate

⁸We have explicitly used the operator identity $[AB, C] = A[B, C] + [A, C]B$ which holds for arbitrary operators.

representation $\overline{R}^9$. This profound statement follows strictly from the Lie algebra Eq. (B.2) by recognizing that the specific structure constants f_c^{ab} directly give rise to the generators operating in the adjoint representation that effectively transforms the index a of $(T_R^a)^\alpha_\beta$. This structurally also means that

$$R \otimes \overline{R} \otimes \text{Adj} = \mathbf{1} \oplus \dots \quad (\text{B.15})$$

Moreover, since the adjoint is physically a real representation (the f_c^{ab} are completely real numbers and thus the group elements $e^{i\alpha_a T_{\text{Adj}}^a}$ are strictly real), one dynamically understands that

$$\text{Adj} \otimes \text{Adj} = \mathbf{1} \oplus \dots \quad (\text{B.16})$$

which perfectly matches with the fact that the established Killing metric δ^{ab} is an invariant tensor critically used to construct scalar products (more generally, the tensor δ_β^α for the arbitrary representations R and $\overline{R}$ is similarly an invariant tensor). Then combining Eq. (B.15) and Eq. (B.16) logically implies

$$R \otimes \overline{R} = \text{Adj} \oplus \dots \quad (\text{B.17})$$

which is physically interpreted by saying that $(T_R^a)^\alpha_\beta$ strictly act as Clebsch-Gordan coefficients: they systematically combine the tensors residing in the representation R with those located in the representation $\overline{R}$ in order to cleanly extract a tensor structurally transforming in the adjoint. Said differently, Clebsch-Gordan coefficients are fundamentally invariant tensors.

Finally, let us meticulously define another important invariant tensor, the totally symmetric d^{abc} tensor, cleanly introduced together with the defining anomaly coefficients $A(R)$ by

$$A(R)d^{abc} = \frac{1}{2} \text{tr} (T_R^a \{T_R^b, T_R^c\}) \quad (\text{B.18})$$

where the overall constant normalization may be conventionally fixed by setting $A = 1$ exclusively for the fundamental representation. It is proven to be totally symmetric and prominently appears in the advanced study of chiral anomalies. The absolute only simple groups that fundamentally have a non-vanishing d^{abc} tensor, and therefore dynamically permit a cubic Casimir operator structurally given by $C_3 \sim d^{abc} T^a T^b T^c$, are $SU(N)$ specifically for $N \geq 3$ and the group $SO(6)$.

Cartan-Weyl basis

It is mathematically often very useful to strategically rewrite the generators characterizing a Lie algebra in the specialized Cartan-Weyl basis. This basis is cleanly defined by first finding the maximal allowable number of generators (or rather, independent linear combinations of said generators) denoted as H_i that strictly commute between themselves:

$$[H_i, H_j] = 0. \quad (\text{B.19})$$

This important maximal number is definitively called the *rank* of the group. They are canonically taken to be hermitian operators, and they rigidly define the Cartan subalgebra of the entire Lie algebra. Since they strictly commute, quantum mechanics dictates they can be diagonalized simultaneously in any given matrix representation, and the resulting eigenvalues are mathematically

⁹One may recall from linear algebra that given a representation R , one definitively finds that $R^{-1,T}$, R^* , and $R^{-1,\dagger}$ are also formally representations. These four mathematically distinct representations acts linearly on vectors $v^\alpha, v_\alpha, v^{\dot{\alpha}}, v_{\dot{\alpha}}$ belonging respectively to the appropriate vector spaces (the original vector space, its rigorous dual, its complex conjugate, and finally the dual complex conjugate). For strict unitary representations, one naturally has $v^{\dot{\alpha}} \sim v_\alpha$ and $v_{\dot{\alpha}} \sim v^\alpha$.

called the *weights*. This specific definition successfully generalizes the familiar angular momentum generator J_3 characteristic of $SU(2)$, which represents a group possessing a rank of exactly 1. J_3 is precisely the designated generator that is usually diagonalized in standard quantum mechanics¹⁰. The particular eigenvalues (weights) calculated specifically for the adjoint representation are uniquely called *roots*.

The remaining generic generators are smartly combined into complex combinations so that they logically correspond one-to-one to the roots α_i :

$$[H_i, E_\alpha] = \alpha_i E_\alpha \quad (\text{B.20})$$

which can be cleanly interpreted by physically saying that α_i strictly serve as eigenvalues while the operators E_α structurally act as eigenvectors (the root α acts as a geometric vector possessing components α_i). The specialized generators E_α inherently cannot be hermitian, but rather one can mathematically show that $E_\alpha^\dagger = E_{-\alpha}$, meaning that if α represents a valid root then necessarily $-\alpha$ is logically also a valid root. They perfectly generalize the $J_\pm$ angular momentum step operators traditionally found in $SU(2)$. Finally, one possesses the remaining set of structure constants that logically appear when explicitly calculating:

$$[E_\alpha, E_\beta]. \quad (\text{B.21})$$

The underlying Jacobi identity can be systematically used to rigorously study them, and, in particular, one fundamentally finds that

$$[E_\alpha, E_{-\alpha}] = \alpha_i H_i. \quad (\text{B.22})$$

which beautifully also generalizes the standard $SU(2)$ case.

This highly structured basis (along with a closely related one technically called the Chevalley basis) proves to be very useful in cleanly deriving the most general properties defining Lie algebras, existing in a close and beautiful analogy with the established theory governing angular momentum in quantum mechanics. In particular, it significantly helps in rigorously constructing the complete mathematical classification enumerating all simple Lie algebras, a towering achievement which is largely due to Killing and Cartan. This elegant classification is visually encoded in the schematic Dynkin diagrams illustrated in fig. 1.

The specific algebras explicitly depicted there logically correspond to the following continuous compact groups: $A_n = SU(n+1)$, $B_n = SO(2n+1)$, $C_n = Sp(2n)$, and $D_n = SO(2n)$, where the integer parameter n precisely denotes the rank. The remaining isolated algebras denoted G_2, F_4, E_6, E_7, E_8 physically correspond to the so-called exceptional groups.

TODO: add image of Dynkin diagrams

¹⁰Recall the foundational $SU(2)$ algebra: $[J_3, J_\pm] = \pm J_\pm$ and $[J_+, J_-] = 2J_3$.